Label Verifier – User Manual

CCL Middle East – Pakistan · Variable Data Verification System

Process diagrams: process-flow.md.

> The screenshots were taken from a demo system with sample data (job cards, codes and users are fictitious).

1. What the system does

Label Verifier checks that every variable-data code (QR code or serial number) printed on a label or carton belongs to the correct production job, and that no code is used twice.

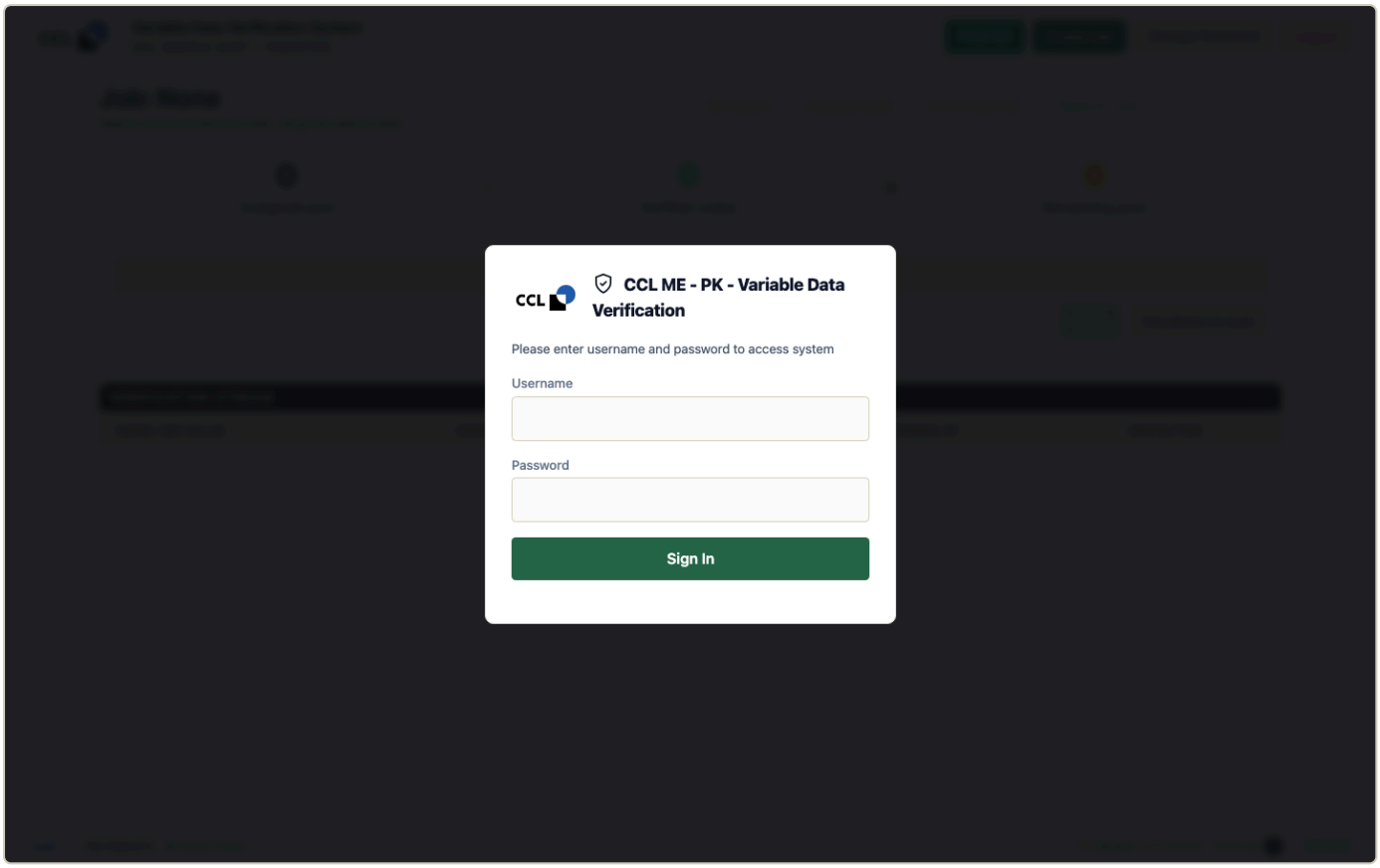
1. A job's code list (CSV/TXT) is uploaded to create a **Job Card**.
2. Operators **scan each printed code**. The system answers PASS, Duplicate or Unknown within a moment.
3. A **Final QC** step checks sampled packed items and prints a certificate.
4. Everything is recorded in reports and an audit trail.

Roles

Role	Can do
QC officer	Sign in, search a job, scan/verify codes, use the phone scanner, view the QC Report and Lifecycle Audit, use Final QC. Can create jobs only if an admin ticked *Allow creating new job cards* for them.
QC manager	Everything an officer can, plus create jobs, **Job Control Center** (change status, purge a job) and the global **Audit Trail**.
Administrator	Everything a manager can, plus **Manage Users** (create users, change roles, reset passwords, suspend accounts).

2. Signing in

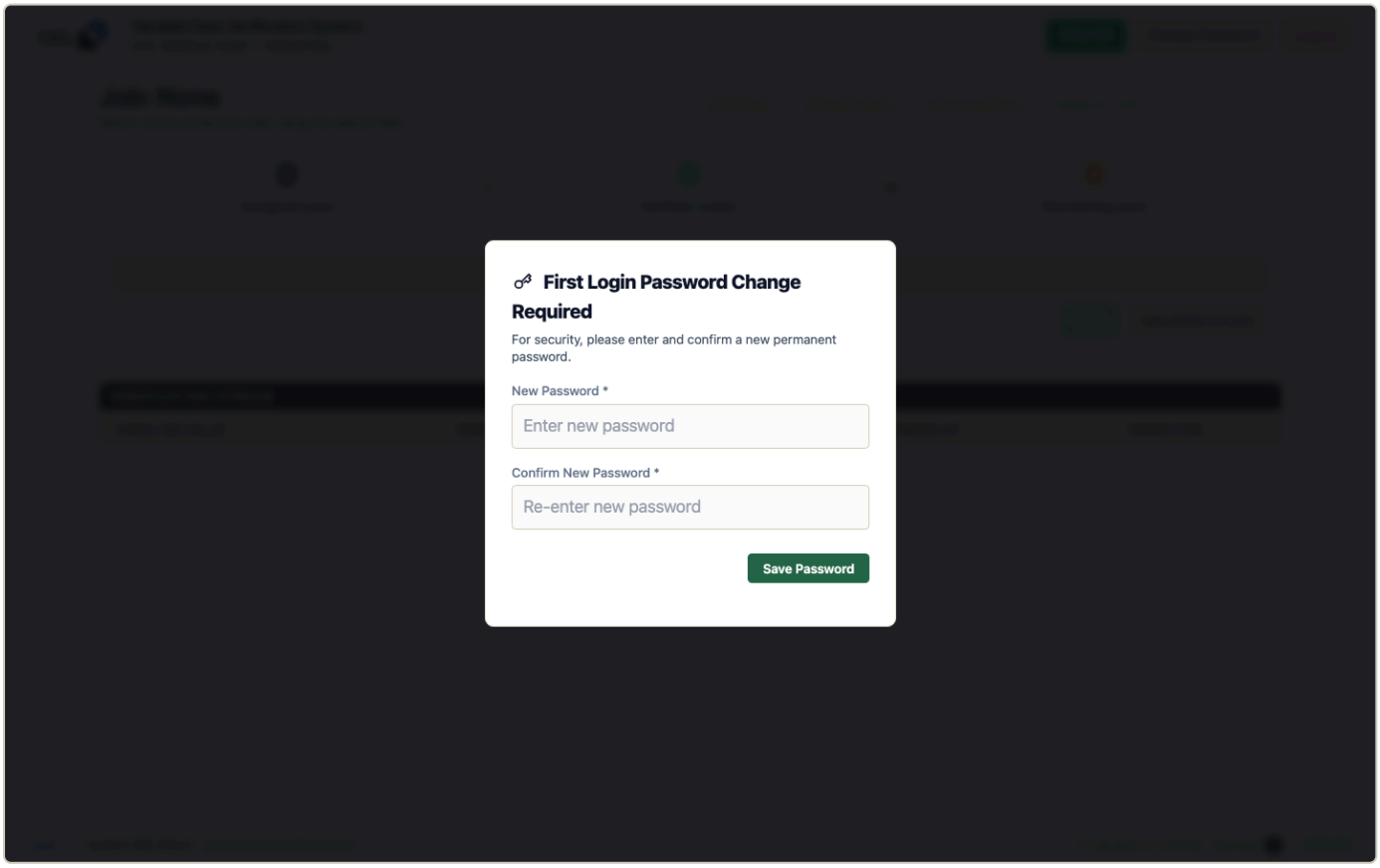
Open the address given by your IT team (for example `https://192.168.1.50`) in Chrome, Edge or Safari. Enter your **username** and **password** and click **Sign In**.



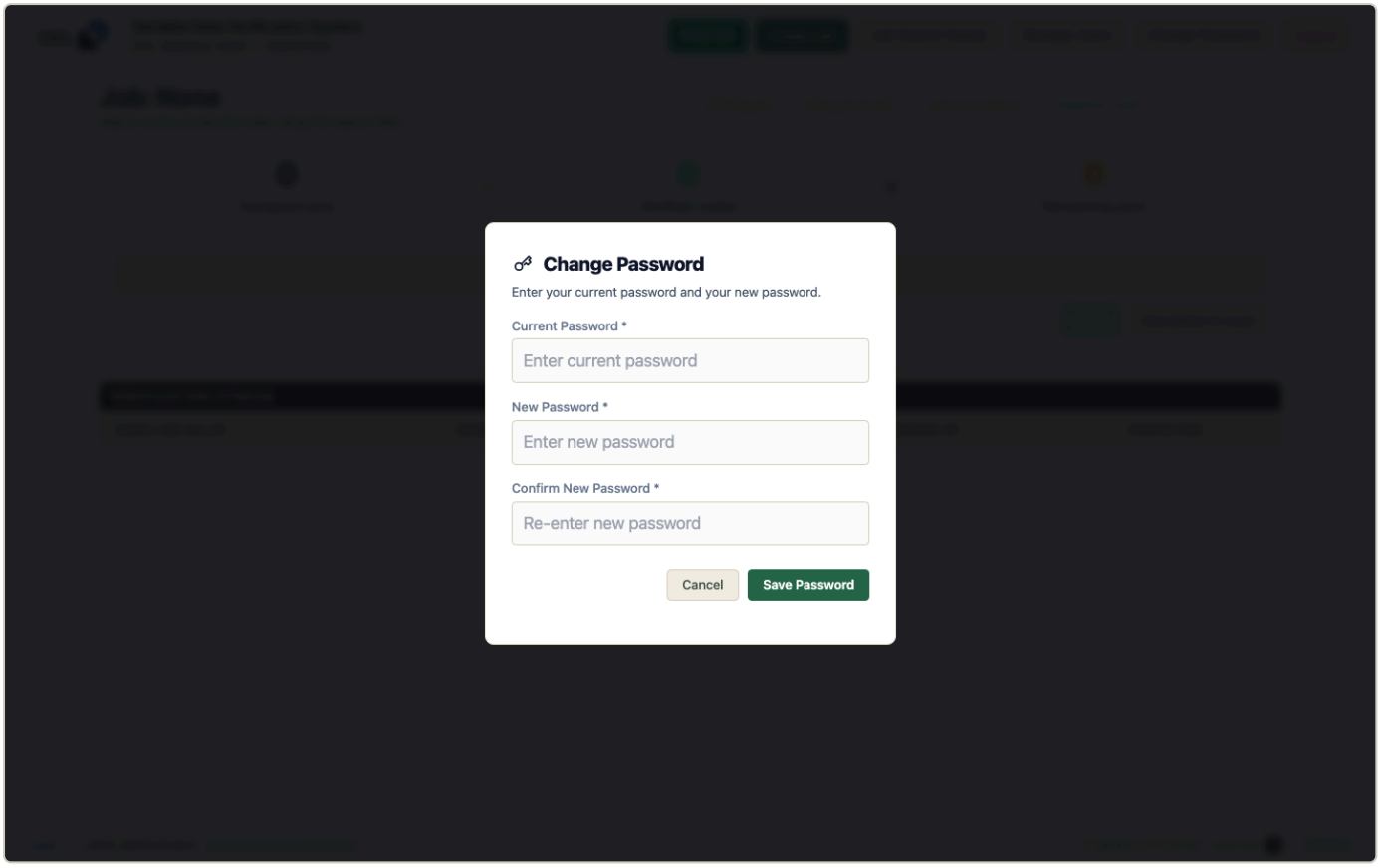
- After several failed attempts the system temporarily blocks further tries (wait a few minutes).
- Passwords must be at least 8 characters.

First login: set a new password

If your password was set or reset by an administrator you must choose a new one before using the system.

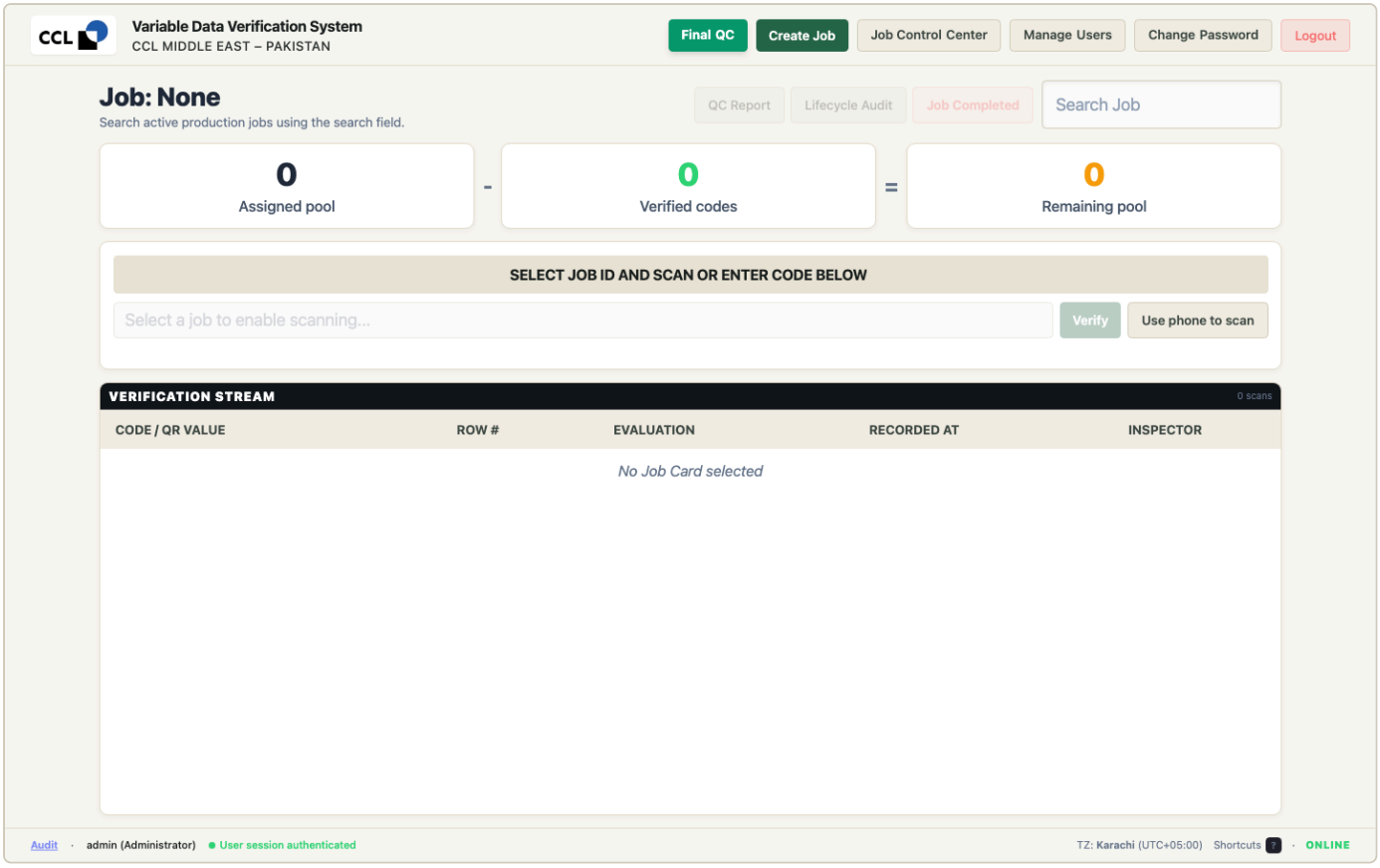


Enter the new password twice and click **Save Password**. You can change it again any time with **Change Password** in the top bar.



---

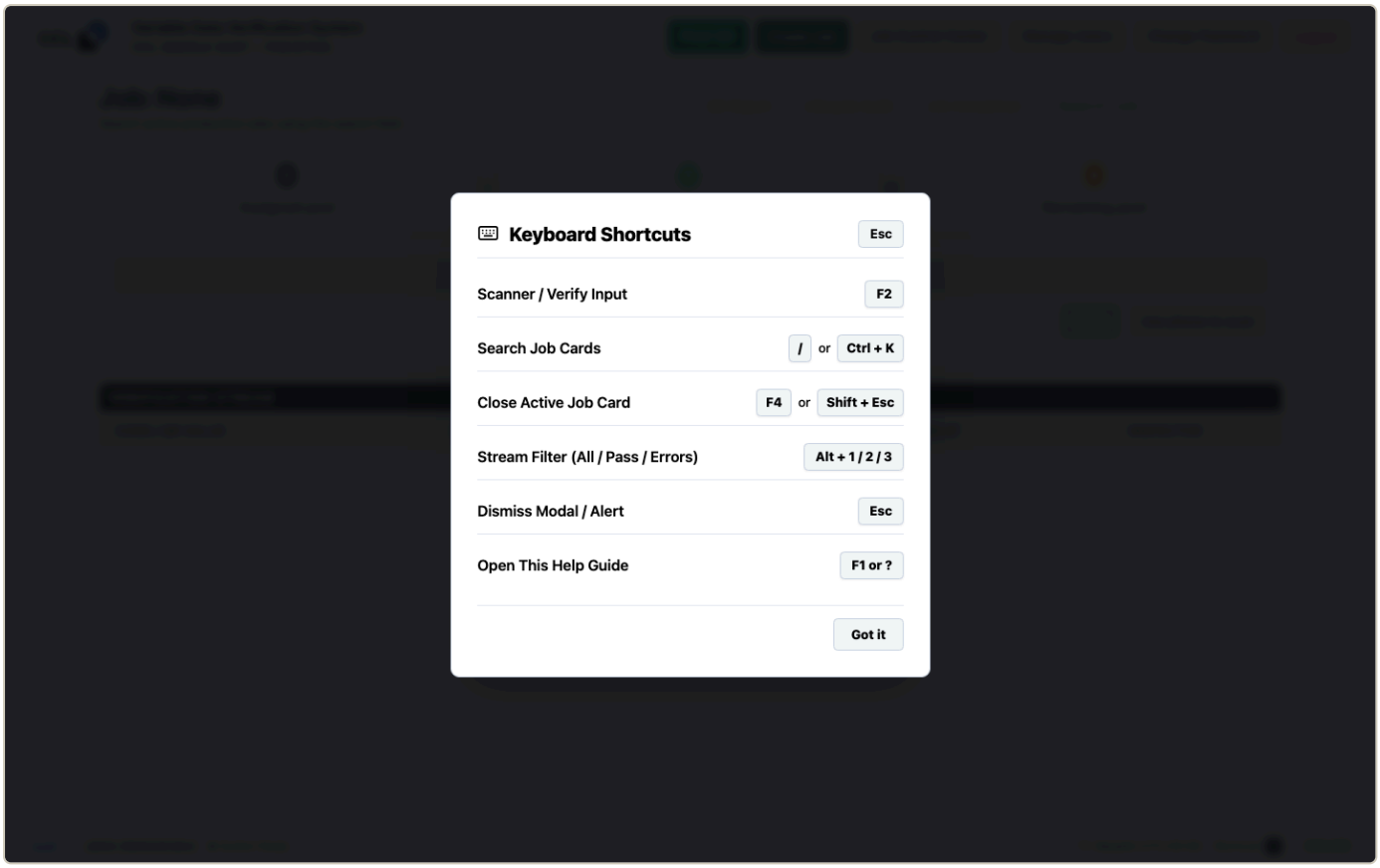
3. The main screen



Area	Purpose
Top bar	**Final QC**, **Create Job**, **Job Control Center**, **Manage Users** (shown according to your role), **Change Password**, **Logout**.
Job header	Shows the selected Job Card. Buttons: **QC Report**, **Lifecycle Audit**, **Job Completed**, and the **Search Job** box.
Counters	**Assigned pool** (codes in the job) – **Verified codes** = **Remaining pool**.
Scan box	Where you scan or type a code, plus **Verify** and **Use phone to scan**.
Verification Stream	Live list of every scan with row number, result, time and inspector.
Footer	Audit link, who is signed in, time zone, keyboard shortcuts, online status.

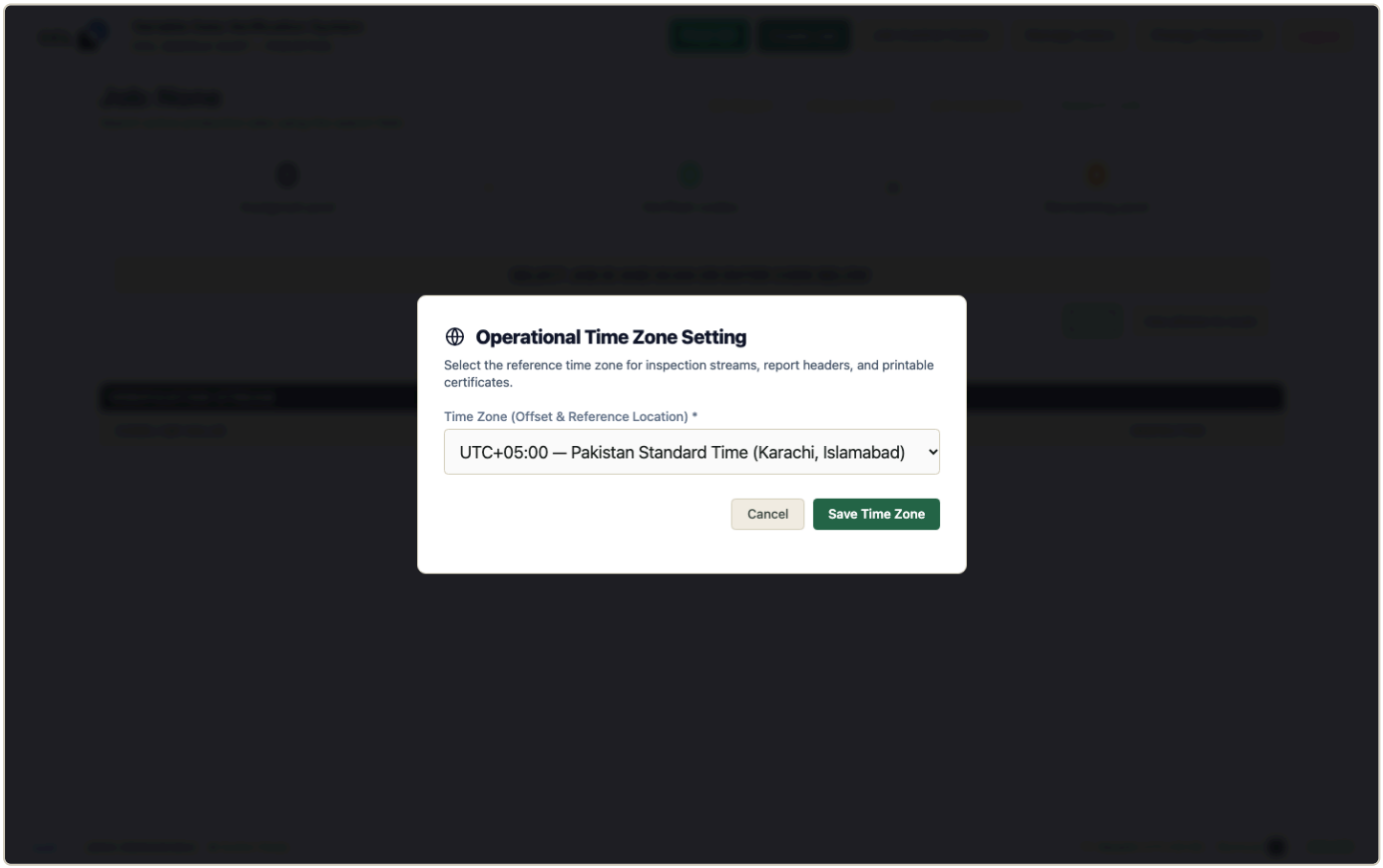
Keyboard shortcuts

Press **F1** or **?** to see them.



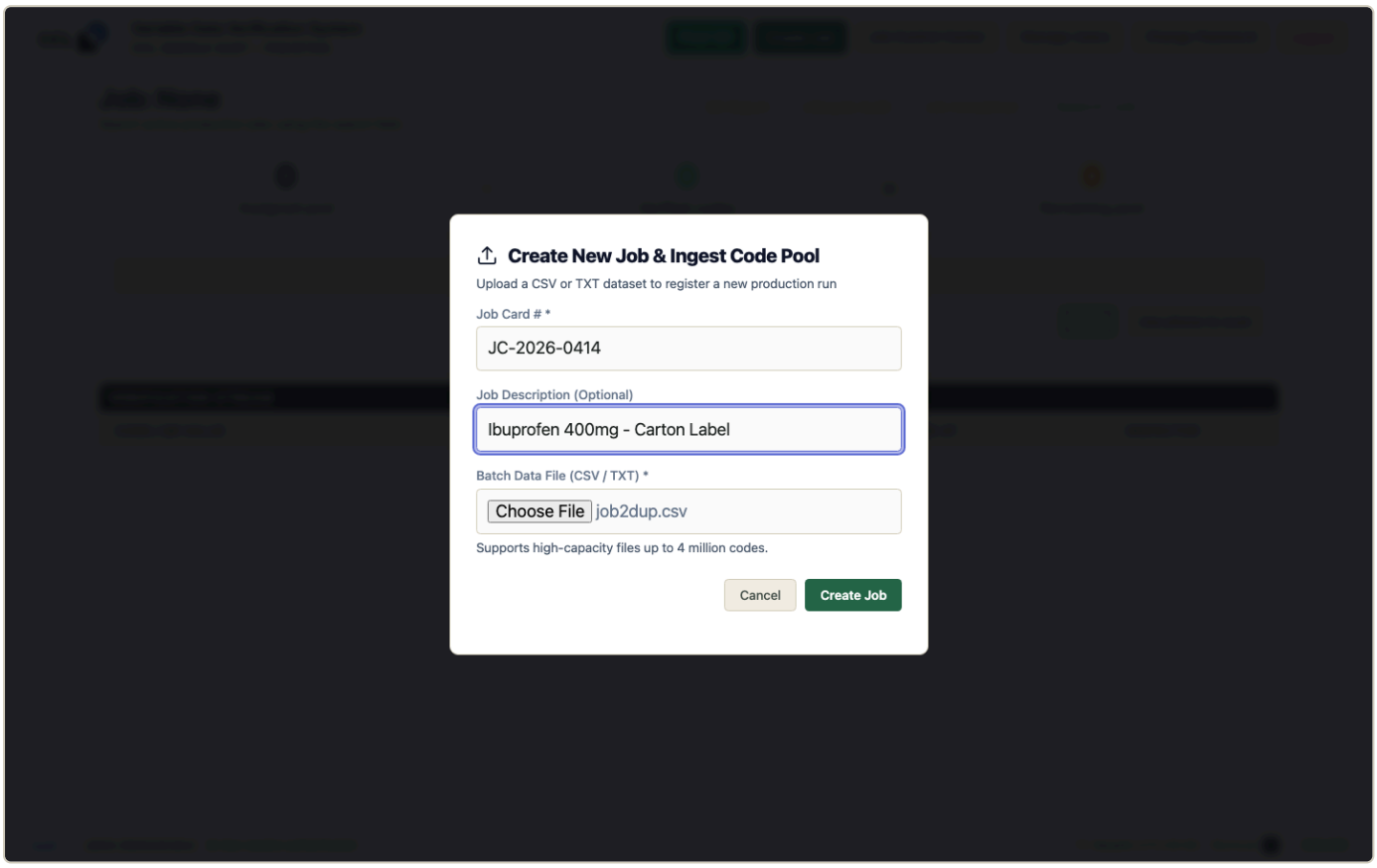
Time zone

Click the **TZ:** label in the footer to choose the time zone used on screens, reports and certificates.



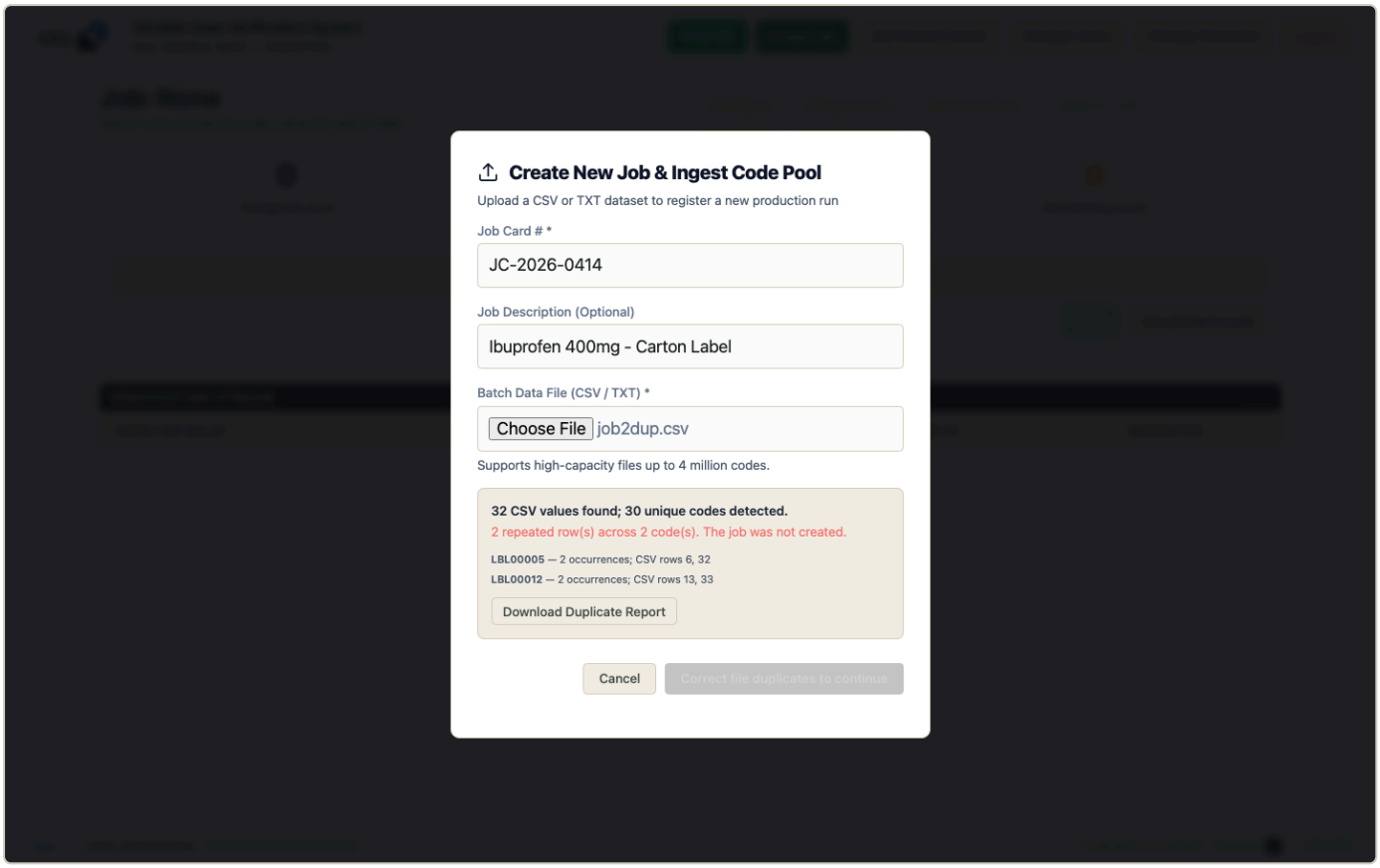
4. Creating a job *(managers, admins, or officers with upload permission)*

1. Click **Create Job**.
2. Enter the **Job Card #** (required) and an optional description.
3. Choose the **Batch Data File** (CSV or TXT; up to about 4 million codes).
4. Click **Create Job**.



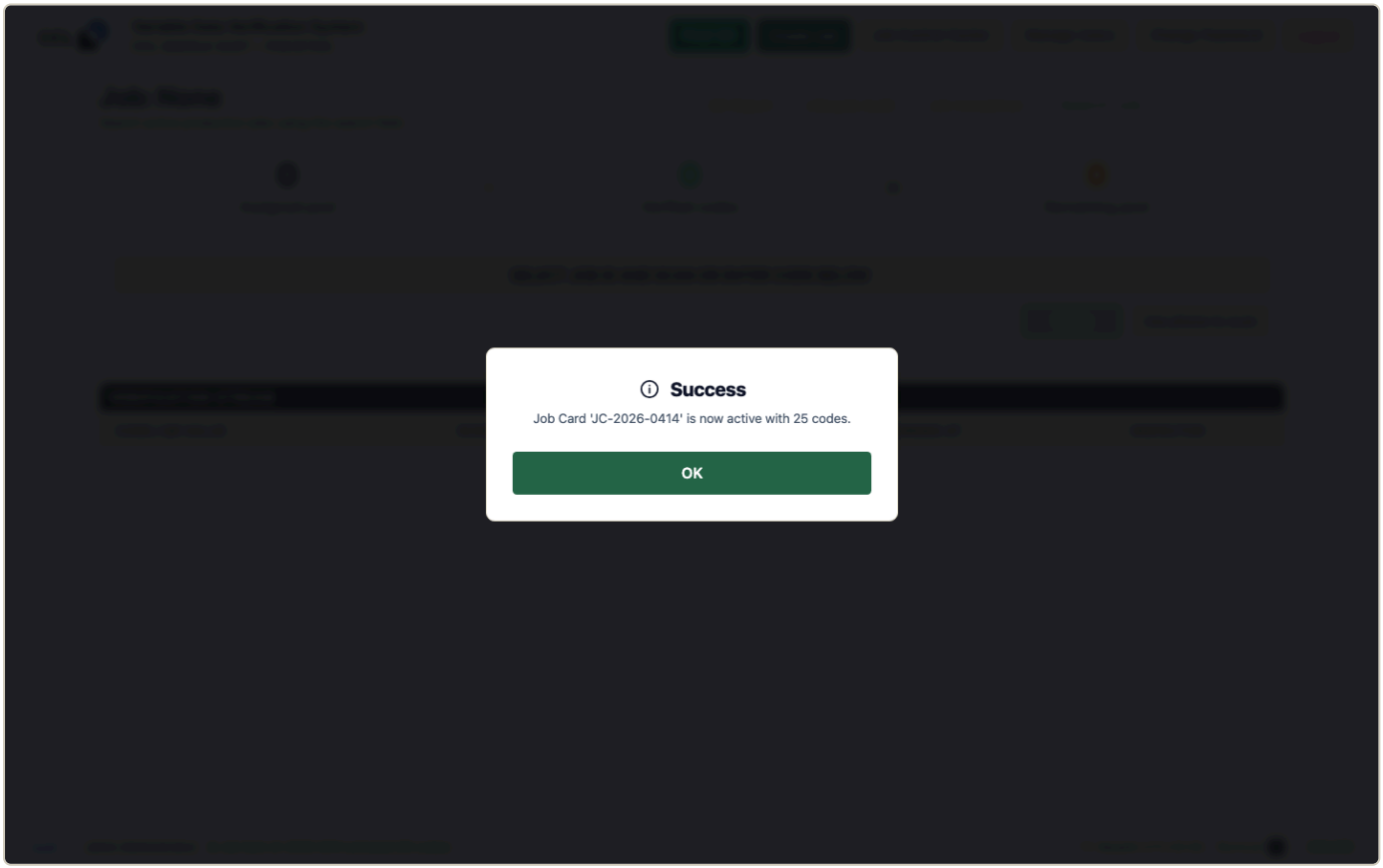
If the file contains repeated codes

The system **refuses** to create a job whose file has the same code more than once. It lists the repeated codes and the CSV rows where they appear. Click **Download Duplicate Report**, fix the source file and upload it again.



Successful creation

Large files are loaded in the background. When finished you see a confirmation and the job becomes **ACTIVE**.



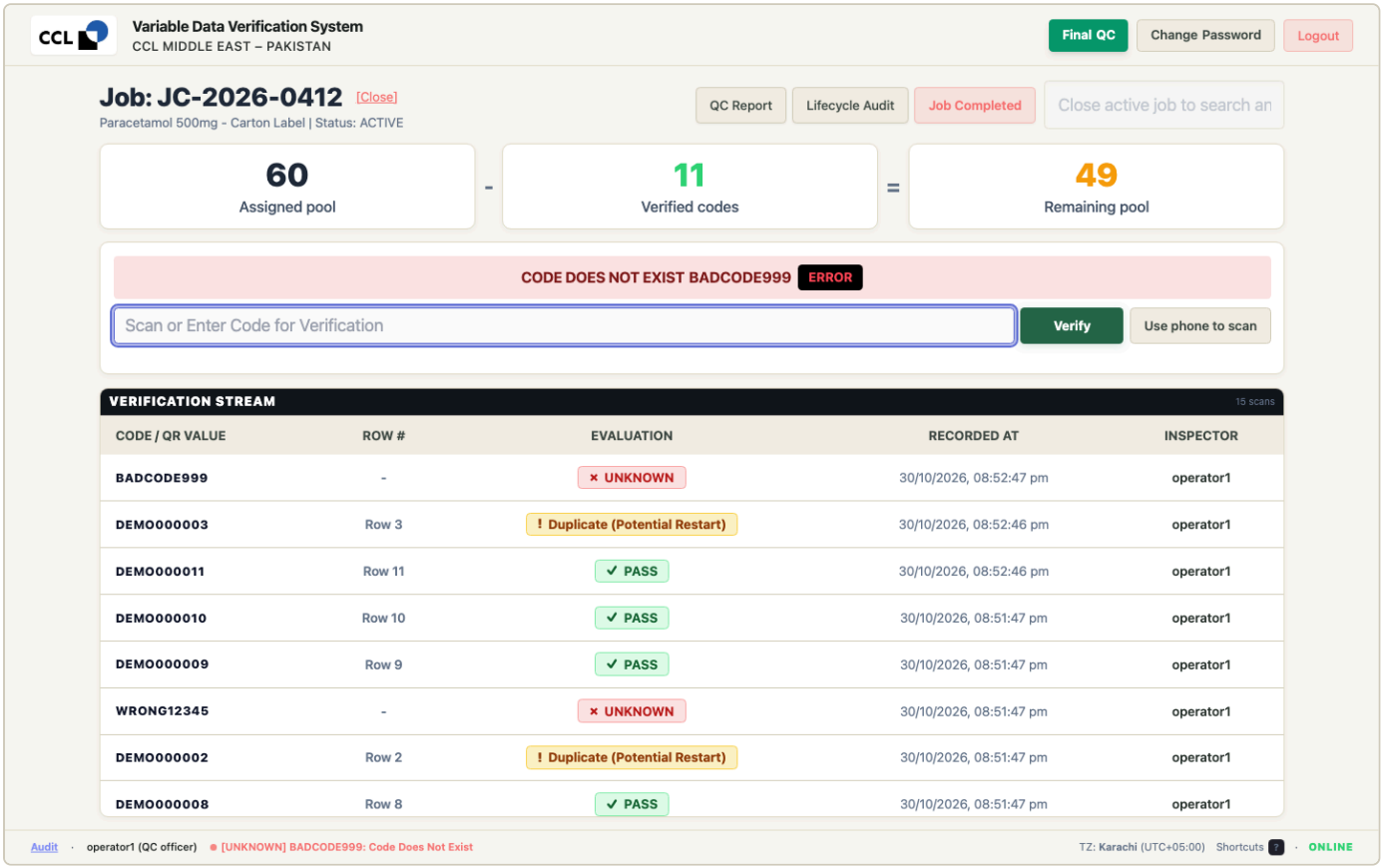
If codes already exist in another job

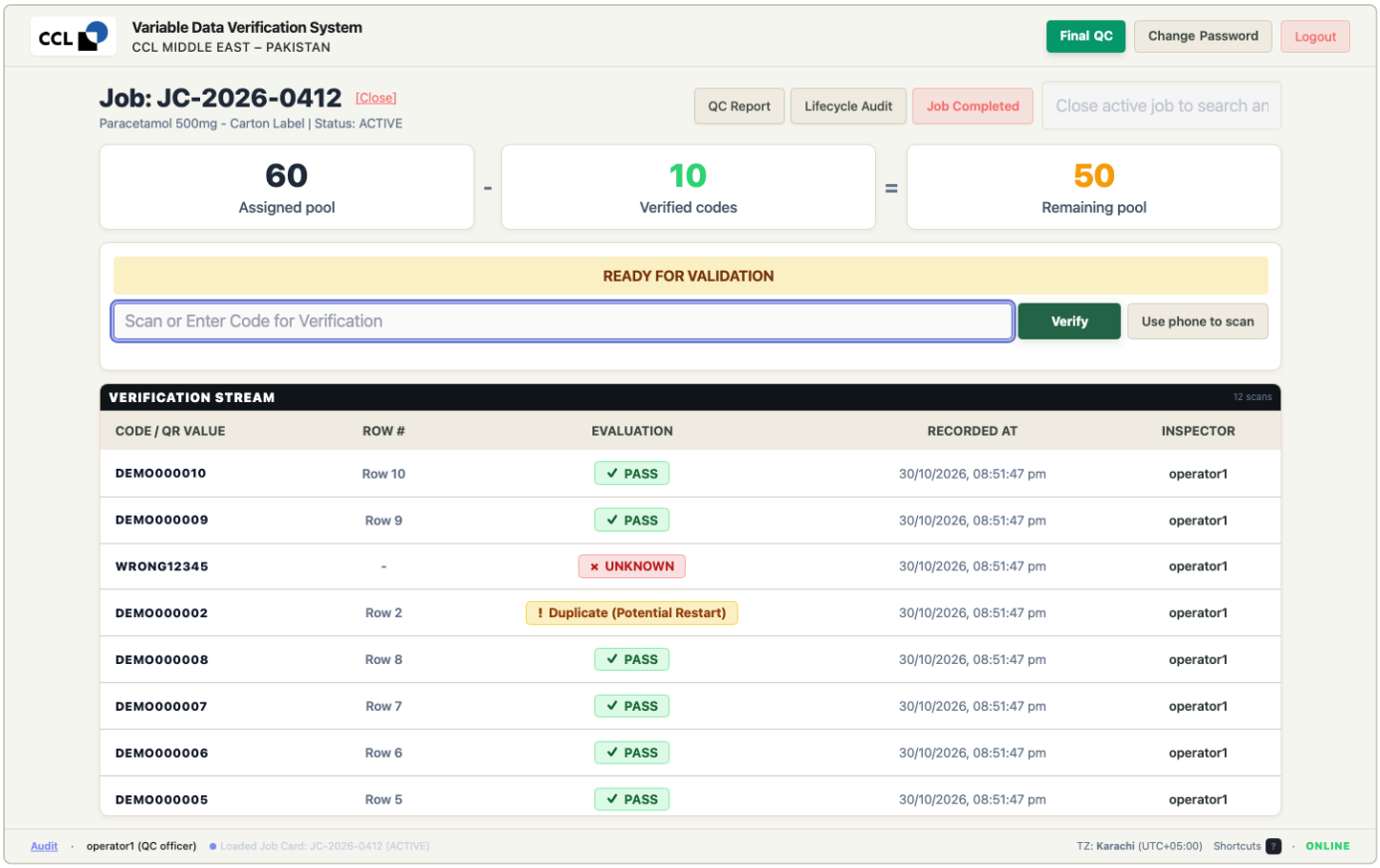
The job stops in **conflict** status and a conflict report is shown. The uploader can either **cancel** the upload or **proceed with a written reason** (optionally skipping the conflicting codes). The reason is stored in the audit trail.

---

5. Verifying codes (daily operation)

1. Click **Search Job** (or press **/**) and choose your Job Card. Only active jobs can be scanned.
2. Click in the scan box (or press **F2**).
3. Scan the label with the keyboard-wedge barcode scanner, or type the code and press **Enter**.
4. Read the result banner and the Verification Stream.





Results

Result	Colour	Meaning	What to do
PASS	Green	Code belongs to this job and had not been used.	Continue.
Pass (Potential Restart)	Amber	Valid, but its row is lower than the highest row already verified (printing may have restarted or run out of order).	Check the sequence of the printed labels.
DUPLICATE / Duplicate (Potential Restart)	Amber	This code was already verified earlier.	Remove or investigate the duplicate label.
UNKNOWN	Red	Code does not exist in this job (wrong job, misprint, or unreadable).	Re-scan, check you selected the right Job Card, escalate if it repeats.
BLOCKED	–	The job is not active (inactive or completed).	Ask a manager to reactivate the job.

Use **Alt + 1/2/3** to filter the stream by All / Pass / Errors. Click **[Close]** next to the job name (or press **F4**) to select a different job.

Finishing a job

When all codes are verified or the run is over, click **Job Completed**. The job stops accepting scans (a manager can reactivate it).

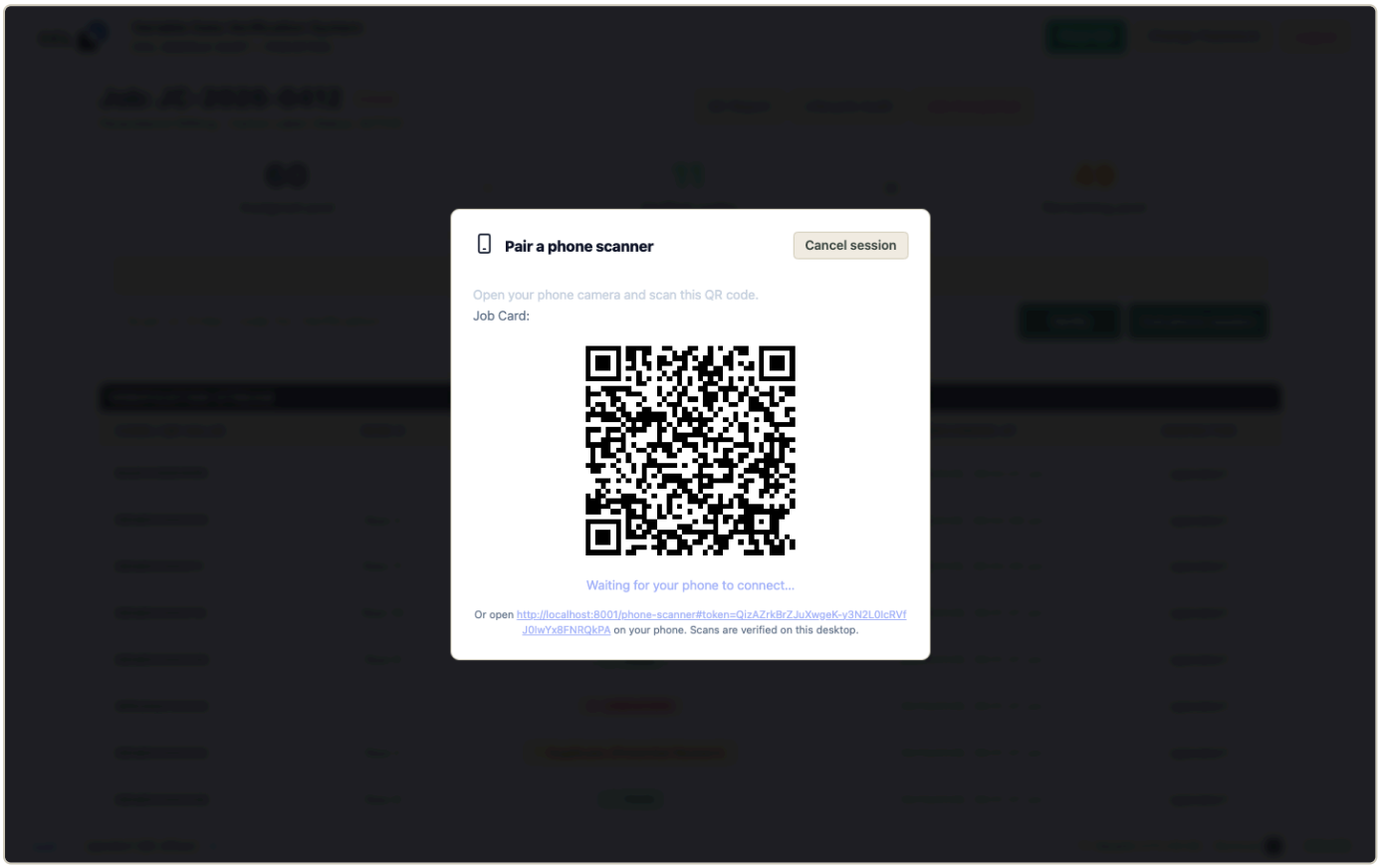
---

6. Phone scanner

Use a phone camera instead of a USB scanner.

Requirements: the phone is on the company Wi-Fi, and the system is opened with its **HTTPS address** (not `localhost`). Your IT team installs the company certificate on phones once.

1. On the desktop select an active job and click **Use phone to scan**.
2. Scan the **QR code** with the phone's camera (or open the link shown).

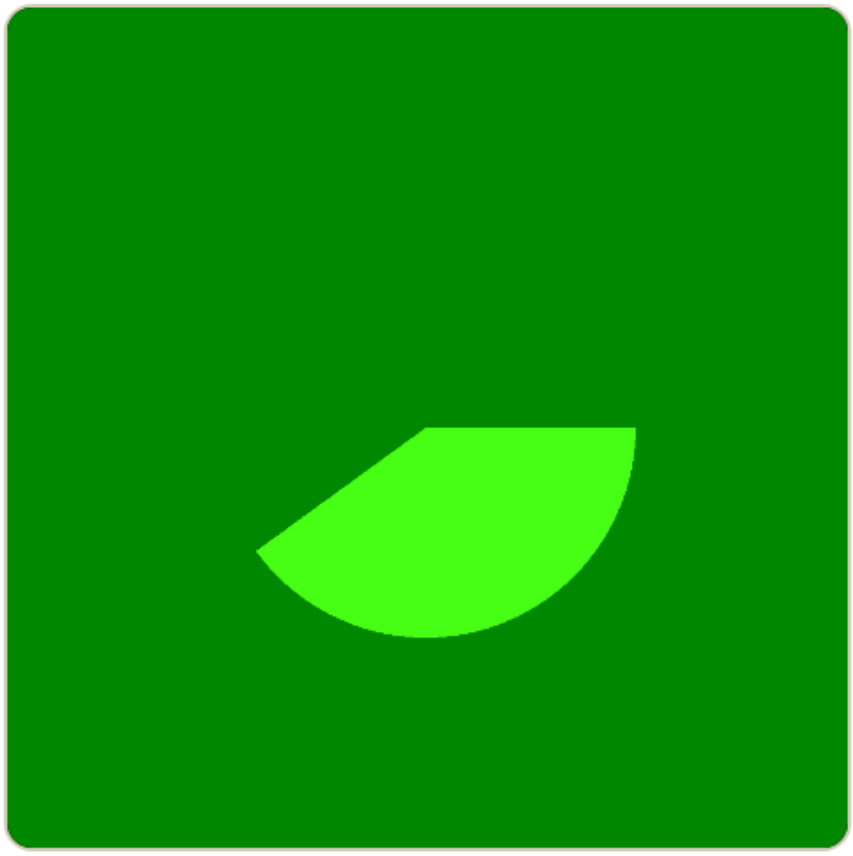


3. The phone page connects to the job and starts the camera. Choose **Barcode / QR** or **Printed text (OCR)**. Use the **1x / 2x / 3x** buttons or slider to zoom.

Phone scanner

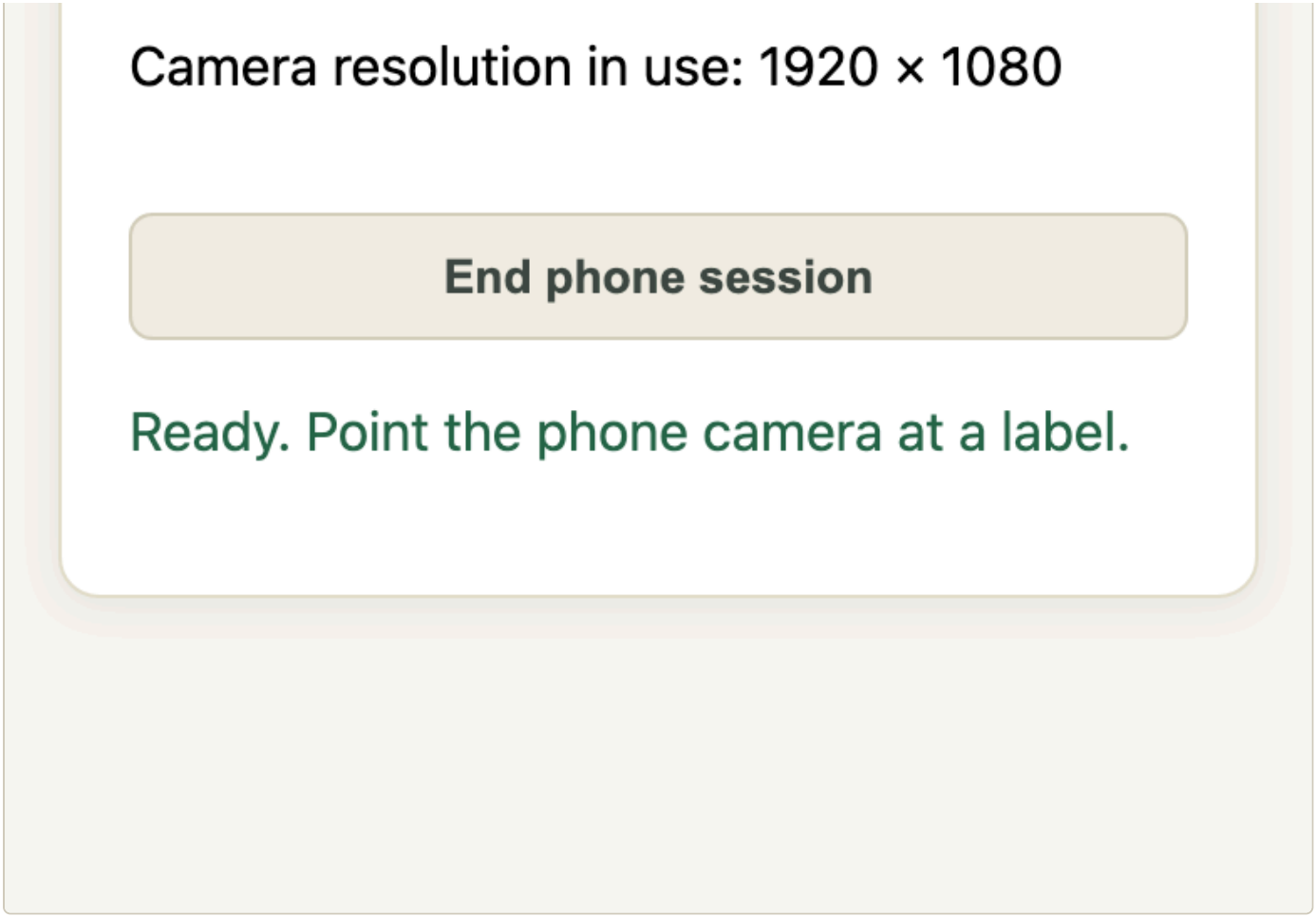
Connected to Job Card JC-2026-0412

Barcode / QR | Printed text (OCR)



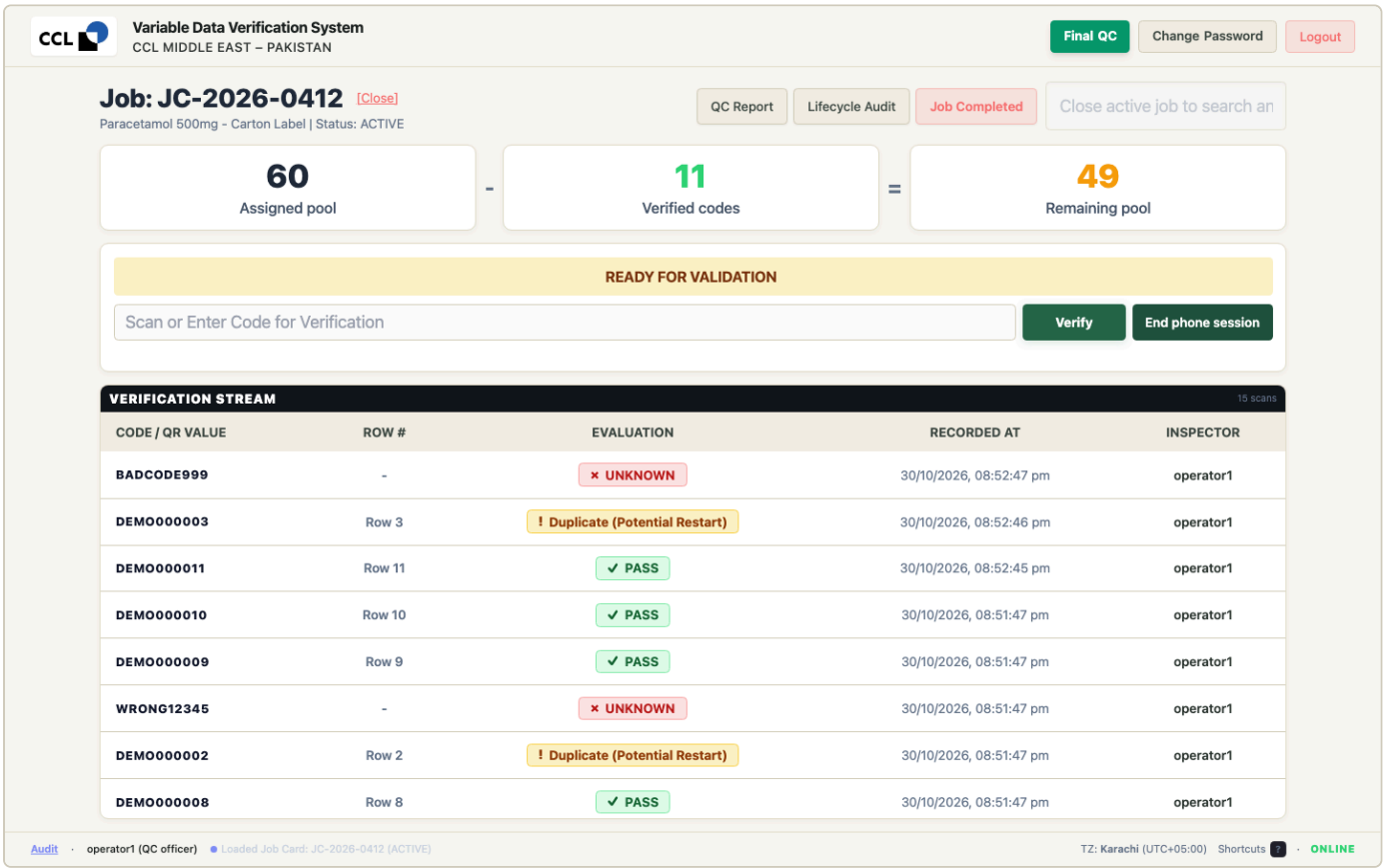
1x 2x 3x

Zoom unavailable on this camera; move closer for a sharper image.

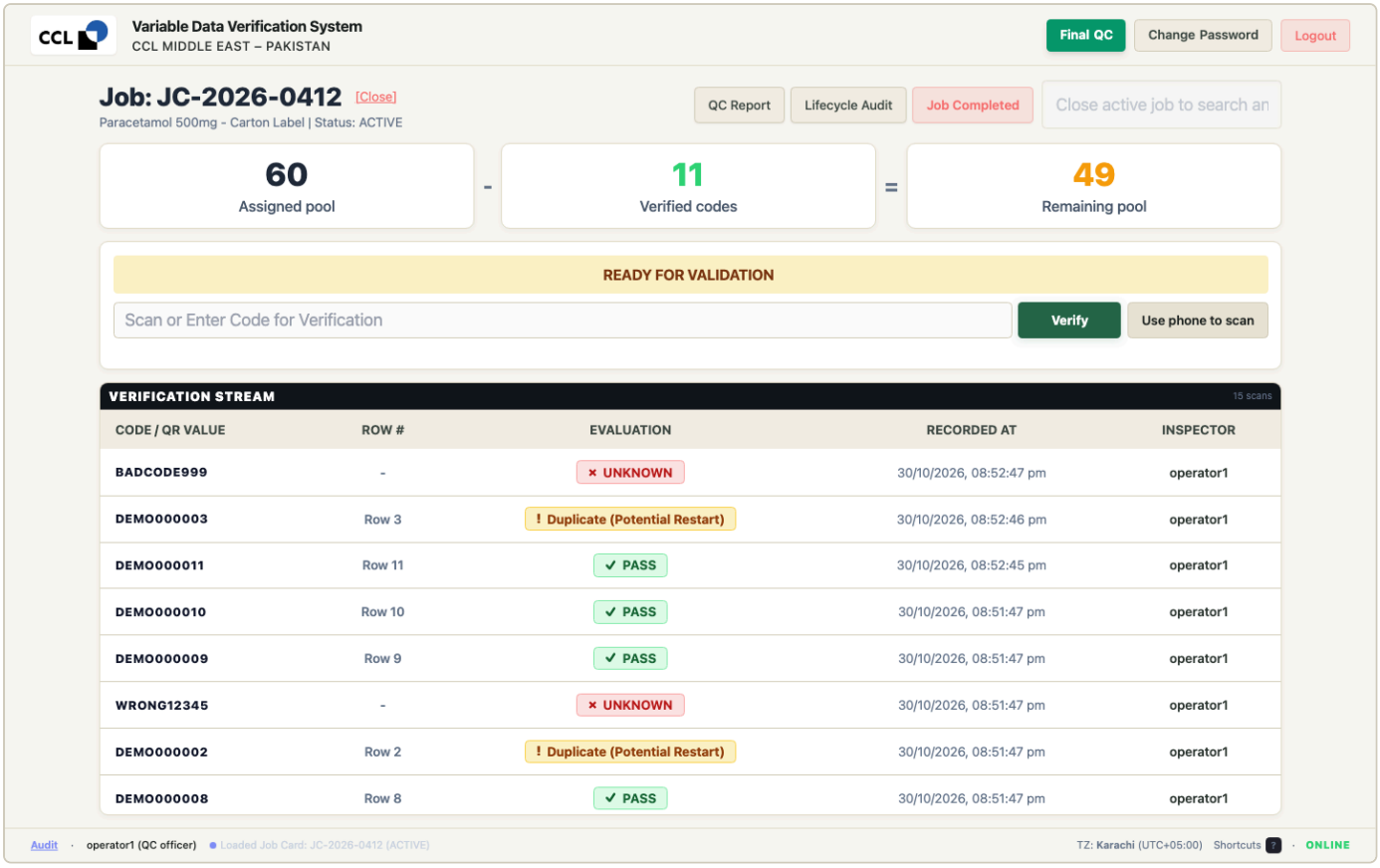


(In this demo the phone shows a test pattern instead of a real camera.)

4. Point the camera at a code. It is sent to the desktop and verified; results appear in the Verification Stream. The pairing window closes by itself.



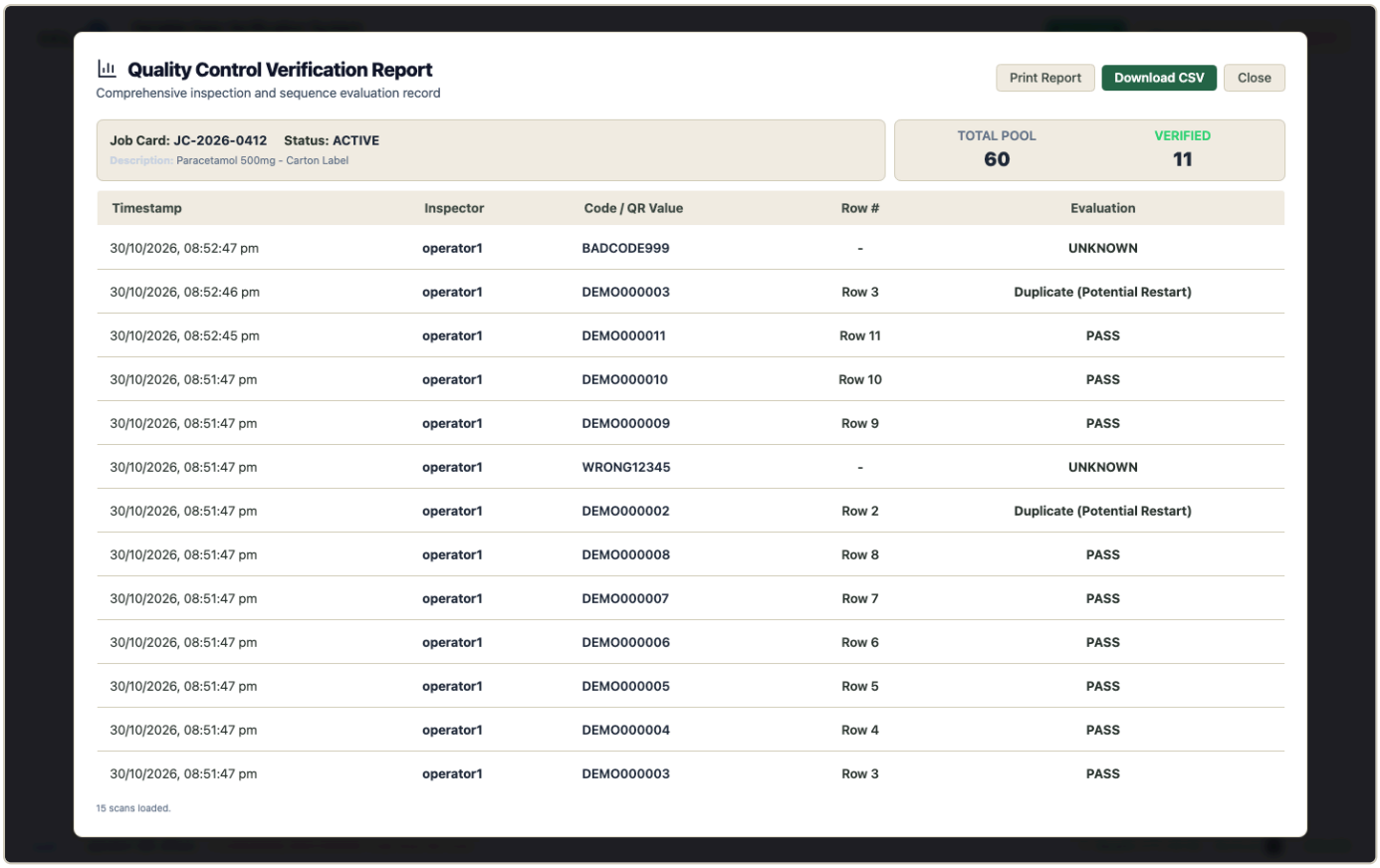
5. To stop, tap **End phone session** on the phone, or **End phone session** on the desktop. **The session also ends automatically when the phone page is closed, the browser is closed, or the phone leaves the page for more than about 30 seconds.** Scan the new QR code to start again.



7. Reports

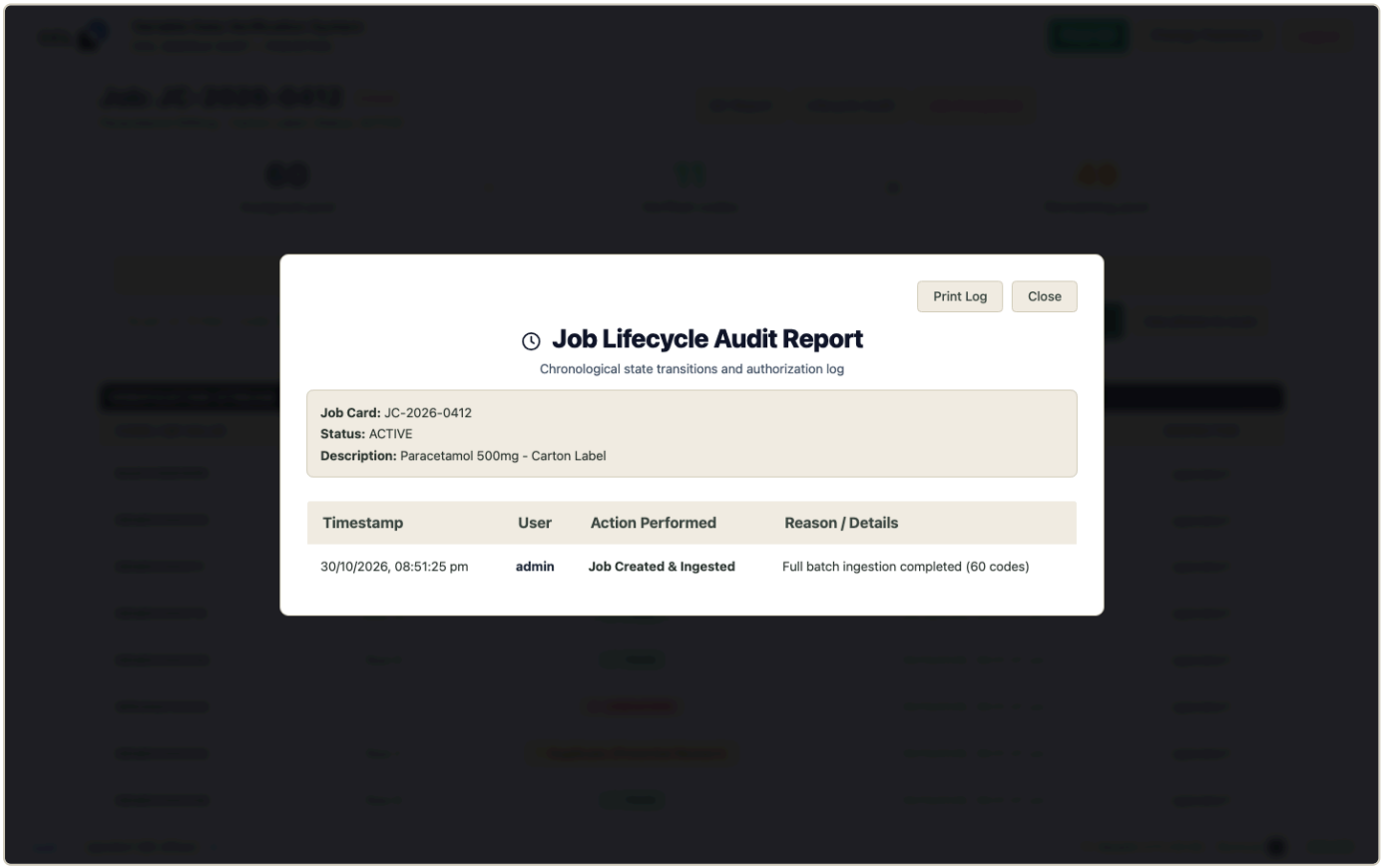
QC Report

Click **QC Report** on the job header: every scan with time, inspector, code, row number and result. Use **Print Report** or **Download CSV**.



Lifecycle Audit

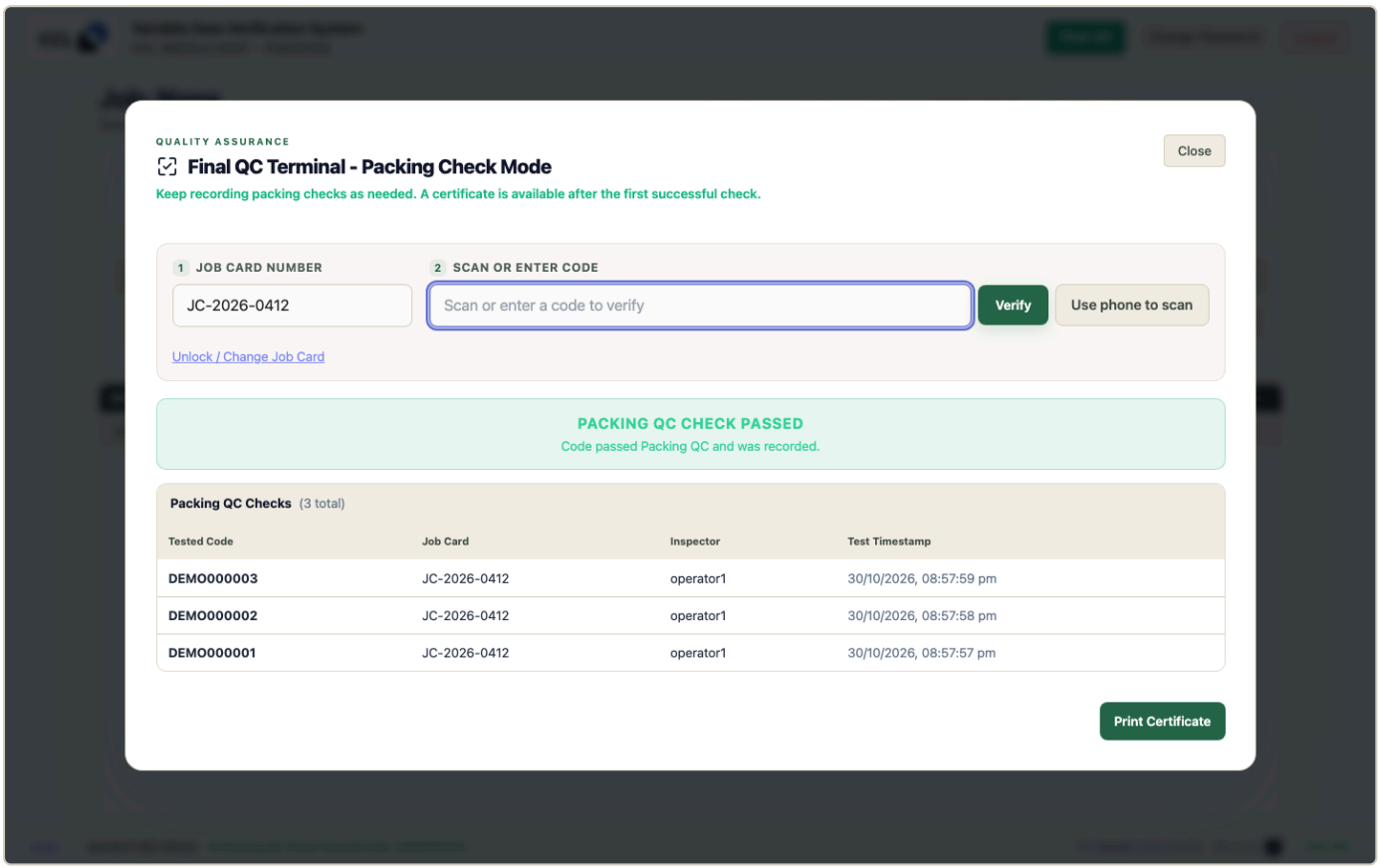
Click **Lifecycle Audit** for the history of the job itself (created, activated, completed, purged) with the user and reason.



8. Final QC Terminal

Used after printing, to check sampled items at packing.

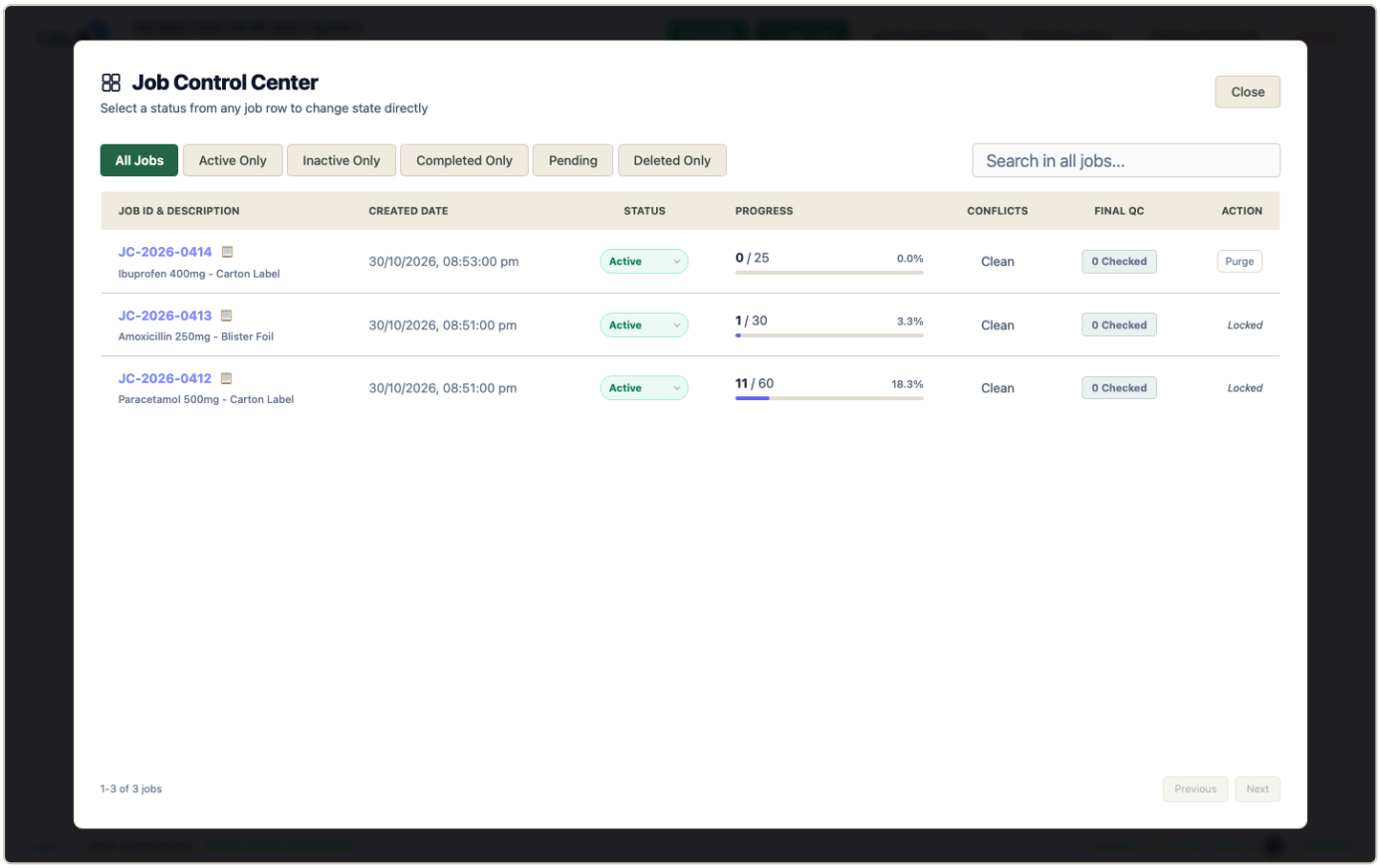
1. Click **Final QC**.
2. Enter or scan the **Job Card number** and press **Enter**.
3. Scan or type the code from the packed item and click **Verify**. Each successful check is added to the list.
4. Click **Print Certificate** (available after the first successful check).



Use **Unlock / Change Job Card** to switch to another job. The phone scanner can also be used here.

9. Job Control Center *(managers and admins)*

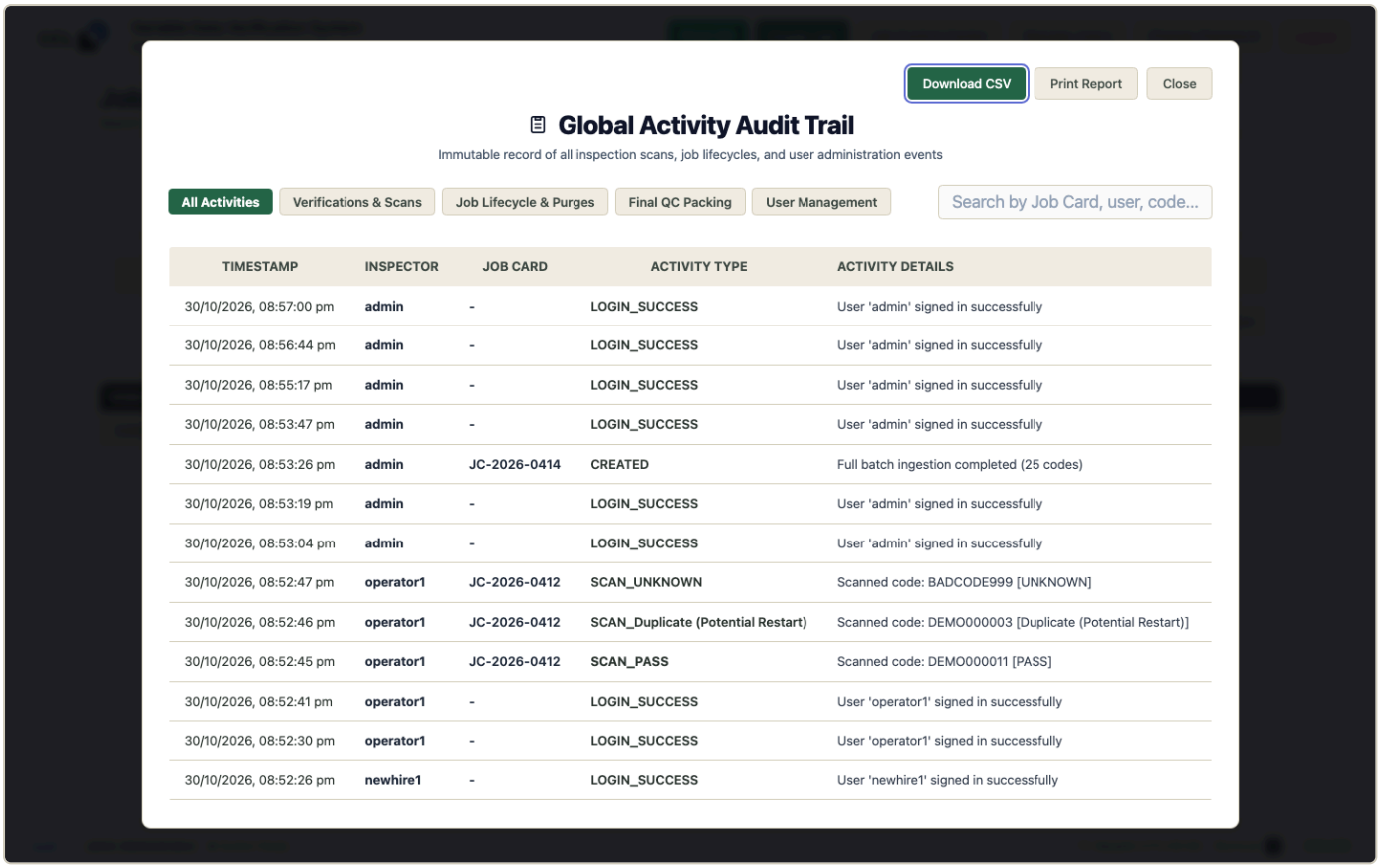
Click **Job Control Center** to see all jobs with date, status, progress, conflicts and Final QC counts.



- Filter by **All / Active / Inactive / Completed / Pending / Deleted**, or search.
- Change a job's status from its status drop-down (you confirm with your password).
- Click a Job ID to open it on the main screen.
- **Purge** permanently removes a job that has no verified codes (requires your password and is recorded in the audit trail). Jobs with scans are locked.

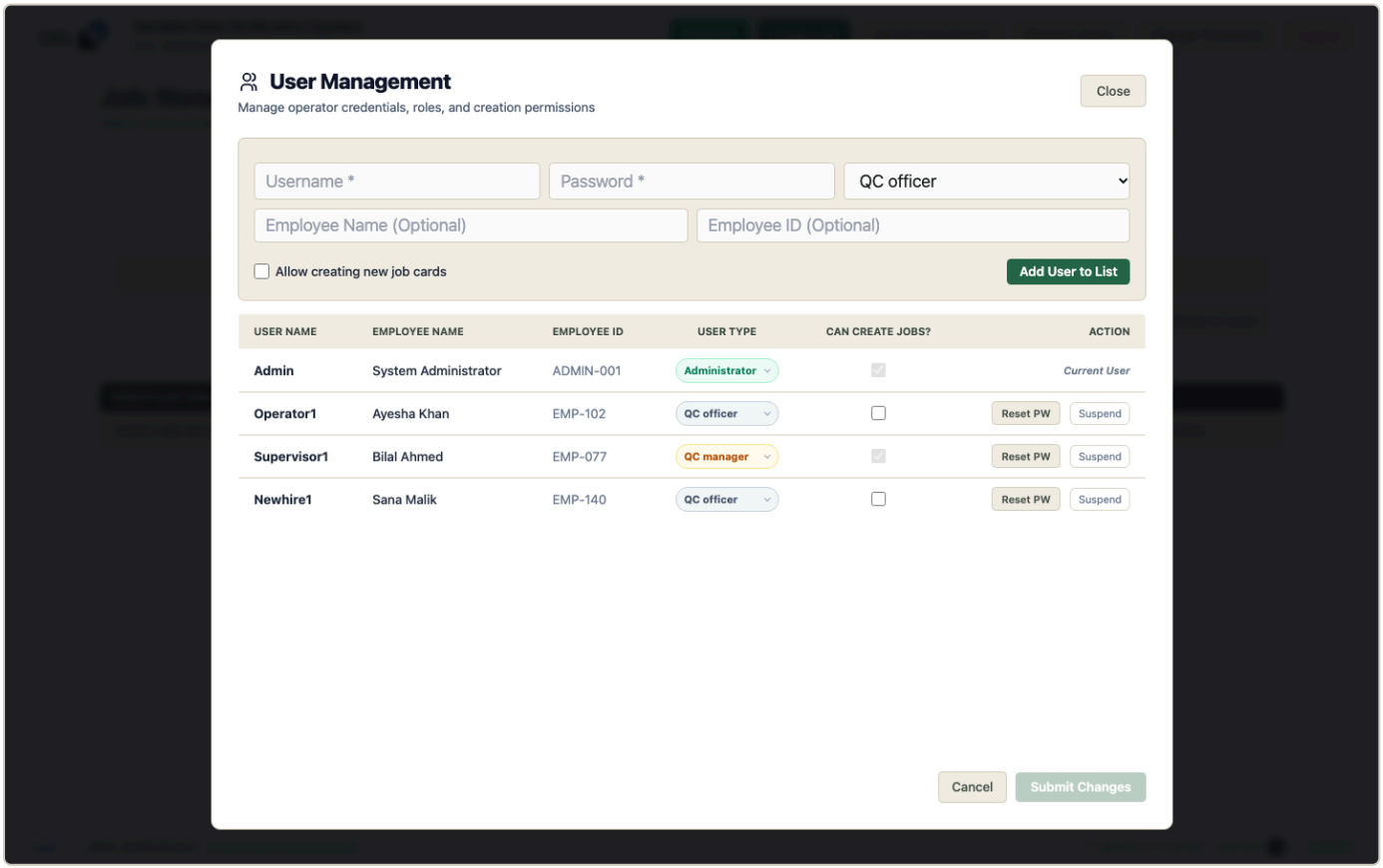
10. Audit Trail *(managers and admins)*

Click **Audit** in the footer. Every login, scan, job change, Final QC check and user-management action is recorded and cannot be edited. Filter by category, search, **Download CSV** or **Print Report**.



11. Managing users *(administrators)*

Click **Manage Users**.



Task	How
Add a user	Enter username, temporary password, role and optional employee name/ID; tick *Allow creating new job cards* if needed; click **Add User to List**.
Change role or upload right	Use the role drop-down and the checkbox in the user's row.
Reset a password	**Reset PW** – the user must choose a new password at next login.
Suspend / re-enable	**Suspend** blocks sign-in without deleting history.
Save	Click **Submit Changes** – nothing is applied until you do.

12. Troubleshooting

Problem	Fix
"Too many attempts" at login	Wait a few minutes, then try again.
Cannot scan: *Job is COMPLETED/INACTIVE*	Ask a manager to set the job to Active.
Everything shows UNKNOWN	Wrong Job Card selected, or the printed code does not match the uploaded file.
Phone says link expired/ended	Start a new phone session from the desktop (links end when the session ends and after 8 hours).
Phone cannot open the page	Same Wi-Fi? Using the HTTPS address, not `localhost`? Company certificate installed?
Camera does not start	Allow camera access in the browser. The page must be HTTPS.
"The server is busy. Please try again in a moment."	Many users at once; wait a few seconds and retry.
Forgot password	Ask an administrator to **Reset PW**.

13. For IT: running and maintaining the system

The application runs as Docker containers (`app` , `db` , `proxy`) on an Ubuntu server. See the project README for installation.

Task	Command (run in the project folder)
Start / update	`docker compose up --build -d`
Status	`docker compose ps`
Logs	`docker compose logs -f app`
Health check	`curl -k https://<server>/healthz`
Backup now	`bash scripts/backup-db.sh`

Task	Command (run in the project folder)
Restore	`bash scripts/restore-backup.sh <file>`

Important settings are in the `.env` file: `SITE_ADDRESS` (the server's IP or DNS name used by browsers and phones), `POSTGRES_PASSWORD`, `SESSION_SECRET`, `BACKUP_DIR` (folder watched by the backup engine) and `BACKUP_RETENTION_DAYS`.